

# Support-gated selective recovery under controlled multi-sensor corruption

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## Abstract

Recovering from sensor corruption requires choosing between a stable base classifier and a recovery endpoint that can both correct and damage decisions. We formulate this choice as controlled-corruption selective recovery. Endpoints and routing thresholds are fitted before evaluation on fixed future batches. The strict test set contains 1,436 observations per mechanism for which a controlled fault reached an available sensor group. The frozen conditional router prevented 94.7% of recovery-induced negative transfers while retaining 9.7% of available corrections. In direct context, always using the base endpoint gives 100% prevention and 0% retention, whereas always using recovery gives 0% prevention and 100% retention. Only five of ten fitted endpoint pairs passed the prospective support gate. The resulting deployment policy reverted unsupported pairs to the base endpoint, achieving 96.2% prevention, 6.6% retention and mean macro-AUROC 0.8132, compared with 0.8115 for the base endpoint. Under equal correction and harm values, calibration-selected all-row and two-stage thresholds yielded utilities of 107.9 and 110.6 net corrections per 10,000 strict observations; unconditional recovery yielded 132.1. At a two-to-one harm cost, two-stage routing was best, but at ratios of five or ten every learned policy had negative test utility and the base endpoint dominated. Severity, sensor-group, mechanism and batch transport changed retention and utility materially. Intervals describe ten repeated endpoint fits on one fixed test set. The evidence supports a reproducible risk-control protocol for controlled corruption, not natural-failure recovery or device-independent field performance.

**Keywords:** multi-sensor fusion; sensor corruption; fault-tolerant classification; selective recovery; negative transfer; cross-fitting

## 1 Introduction

Scientific and industrial systems increasingly combine several channels whose availability and fidelity can change after deployment. Missing inputs, delayed diagnostics and distribution shifts can invalidate a model even when its architecture performs well on a static benchmark [1–4]. This problem is distinct from ordinary random missingness because a faulty sensor can remain present and produce plausible but biased, clipped, noisy or unresponsive measurements. A robust fusion rule must therefore decide not only whether a channel exists, but how strongly it should influence the prediction.

Several training strategies improve tolerance to absent or inconsistent sources. Modality dropout, masking and self-distillation expose models to missing inputs [5–7]. Representation refinement, gradient decoupling and expert routing address source imbalance or train–test mismatch [8–11]. Reliability-aware fusion additionally uses predictive uncertainty, learned credibility

or rejection [12–17]. Quality-aware multimodal fusion (QMF) provides a published failure-aware comparator by deriving modality confidence from predictive energy and aligning confidence with historical loss trajectories [18]. These approaches establish useful design patterns, but they do not make an arbitrary weighting response physically calibrated or comparable across sensors.

Sensor-fault diagnosis has a complementary engineering literature. Redundant sensor networks can localize offset, drift, noise, peaks and stuck responses, and can distinguish local sensor faults from global system changes [19–21]. Hydraulic condition-monitoring studies further show the value of heterogeneous pressure, flow, temperature, power and vibration measurements for component-state classification [22–25]. These studies motivate physically defined sensor groups and independent condition-monitoring validation rather than arbitrary computational groupings alone.

Most open benchmarks nevertheless encode sensor failure through simulation, laboratory intervention or post hoc feature corruption. A recent public AHU resource instead provides one year of building-management-system records from an office, an auditorium and a hospital, with return- and supply-air temperature sensor faults annotated by experienced HVAC engineers [26, 27]. These labels create a useful external field test, although they are rule- and expert-adjudicated operational annotations rather than maintenance-confirmed hardware replacement events.

Recent AHU studies show that the direct diagnosis problem differs from fault-tolerant recovery. Transformer models can exploit long temporal dependencies within a continuously operating hospital, while attention-based transfer and temporal-contextual learning explicitly address building shift and limited labels [28–30]. These architectures preserve fault signatures to identify the failure itself. In contrast, fault-tolerant classification seeks to suppress corrupted evidence while retaining another target, so superiority in one task need not transfer to the other.

We study paired-degradation responsive fusion (PDRF). “Responsive” denotes an operational change in fusion weight after a controlled sensor degradation; it does not imply physical health, calibrated risk or sensor variance. Each training observation is paired with a view in which one available sensor group is degraded. The model learns a bounded response, and directional losses encourage the affected sensor’s response to rise and its fusion weight to fall. The formulation does not invoke an identifiable causal counterfactual. Predictive calibration is evaluated separately because discrimination, uncertainty and calibration are different properties [31–37].

The study makes four contributions. First, it defines a strict recovery estimand that includes only controlled interventions that reach an available sensor group and separates prevention, retained correction and selection denominators. Second, it distinguishes a correctness-disagreement conditional score from the deployment estimand required on all observations. An all-row multinomial router, a two-stage disagreement-preference router, random routing, scalar confidence/entropy rules, a logit-margin rule and a shallow tree provide direct controls. Third, it separates lexicographic safety thresholds from thresholds that maximize a declared calibration-set utility, and it makes a prospective support gate with mandatory base-endpoint fallback the primary deployment policy. Fourth, it evaluates selector transport across unseen mechanisms, calibration mixtures, corruption severities, affected sensor groups and highly imbalanced acquisition batches, while reporting fitting variation and engineering cost. The title and claims are limited to controlled corruption. Hydraulic results are single-rig sensitivity evidence, and expert-annotated air-handling-unit diagnosis remains a negative task boundary because it lacks an independent downstream recovery target.

## 2 Related work

### 2.1 Missing and unreliable sensor inputs

Missing-data mechanisms affect the interpretation of any robustness experiment [38]. Random sensor dropout tests tolerance to absent channels, whereas the present study targets channels that remain available but become corrupted. Corruption augmentation and consistency distillation can improve this setting, but they can also make an apparently novel fusion rule indistinguishable from robust training. Our comparator design therefore gives the main gates identical paired degradations, labels, minibatches and additional forward passes, and separately evaluates early and uniform fusion with clean distillation and modality-dropout self-distillation.

### 2.2 Bounded operational responses and probability calibration

Input-dependent losses can represent aleatoric variation under valid probabilistic assumptions, whereas PDRF uses its bounded response only for weighting [31, 32]. Response ordering and isotonic audits therefore do not convert it into physical variance. Probability quality is assessed separately with NLL, Brier score, multiple ECE definitions and temperature scaling because calibration depends on both multiclass structure and binning [37, 39, 40]. Entropy weighting, deep ensembles and conformal sets provide uncertainty-oriented comparators, not device-health estimates [33, 41]. Cross-building AHU controls likewise distinguish source-only GroupDRO from target-feature adaptation by CORAL or target normalization [42–44].

### 2.3 Selective prediction and model fallback

Selective classification and reject-option learning trade coverage for lower predictive risk [45–47]; recent work additionally studies selection under distribution shift [48]. Learning-to-defer methods route difficult cases to a human or another expert whose error pattern may differ from that of the primary predictor [49, 50]. Classifier cascades explicitly trade predictive accuracy against evaluation cost, while dynamic classifier and ensemble selection estimate which member is locally competent [51, 52]. Mixture-of-experts routing provides a related differentiable allocation mechanism [14]. The present fallback problem is narrower: both endpoints are fixed models of the same sensor-fusion task, no observation is rejected, and the selector specifically controls recovery-induced negative transfer while retaining corrections. Its operating curve is therefore prevention versus retained recovery, rather than conventional risk versus coverage, human deferral utility or average cascade cost.

### 2.4 Evidence and statistical estimands

Classifier comparisons can change with the resampling unit and evaluation estimand [53, 54]. Optimization seeds describe repeated fitting, whereas sample bootstrap intervals describe a fixed set of fitted predictors [55]. McNemar’s paired test evaluates discordant hard decisions on the same observations [56]. We report these levels separately and do not treat observations from one instrument as independent devices. Precision–recall metrics are included because class imbalance can make ROC summaries incomplete [57].

### 3 Methods

#### 3.1 Problem definition

For observation  $i$  and sensor group  $m$ , let  $\mathbf{x}_{im} \in \mathbb{R}^d$  be the feature vector,  $a_{im} \in \{0, 1\}$  the availability indicator,  $q_{im} \in (0, 1]$  an optional diagnostic-quality covariate and  $y_i \in \{1, \dots, C\}$  the target. PDRF estimates

$$p(y_i \mid \{\mathbf{x}_{im}, a_{im}, q_{im}\}_{m=1}^M). \quad (1)$$

When no quality metadata are available, all  $q_{im}$  equal one. Every evaluated observation contains at least one available group. If all groups are unavailable, the implementation abstains before inference because normalized fusion is undefined.

For the routing audit, let  $b_i$  indicate that the base endpoint is correct,  $r_i$  that the recovery endpoint is correct and  $s_i$  that the router emits the recovery output. Table 1 fixes the event terminology used throughout.

Table 1: Hard-decision event definitions and denominators.  $N_C$  and  $N_H$  count correction and harm opportunities;  $n_C$  and  $n_H$  count retained corrections and realized harms. Prevention is  $1 - n_H/N_H$ , whereas recovery retention is  $n_C/N_C$ .

| Quantity                    | Event definition                 | Denominator symbol                                        | Reported rate         |
|-----------------------------|----------------------------------|-----------------------------------------------------------|-----------------------|
| Correction opportunity      | $\neg b_i \wedge r_i$            | $N_C = \sum_i \mathbb{1}(\neg b_i \wedge r_i)$            | $n_C/N_C$             |
| Retained correction         | $\neg b_i \wedge r_i \wedge s_i$ | $n_C = \sum_i \mathbb{1}(\neg b_i \wedge r_i \wedge s_i)$ | events / observations |
| Harm opportunity            | $b_i \wedge \neg r_i$            | $N_H = \sum_i \mathbb{1}(b_i \wedge \neg r_i)$            | $1 - n_H/N_H$         |
| Realized harm               | $b_i \wedge \neg r_i \wedge s_i$ | $n_H = \sum_i \mathbb{1}(b_i \wedge \neg r_i \wedge s_i)$ | events / observations |
| Recovery endpoint selection | $s_i$                            | $N = \sum_i 1$                                            | $\sum_i s_i/N$        |
| Net correct change          | $n_C - n_H$                      | $N$                                                       | events per 10,000     |

#### 3.2 Architecture and bounded weighting

Each sensor group is encoded by affine layers  $d \rightarrow 64 \rightarrow 32$ , with a ReLU after each hidden affine transformation and no normalization or dropout. A biased linear  $32 \rightarrow C$  output produces group-specific logits  $\boldsymbol{\mu}_{im}$ . The response head is exactly Linear(32, 16, bias = true), ReLU, then Linear(16, 1, bias = true). There is no activation after the final linear layer, so its raw response  $u_{im}$  can be positive or negative. The bounded operational response is

$$\begin{aligned} r_{im} &= B \tanh(u_{im}/B), \\ -B &< r_{im} < B. \end{aligned} \quad (2)$$

where the prespecified main setting uses  $B = 3$ . This layer placement matches the released implementation and gives the symmetric open interval in Eq. (2). Across 17,650 clean and faulted seed-observation responses from 10 fitted RO-PDRF-Full models, the observed float32 range was  $-2.899454$  to  $3.000000$ ; the latter is numerical rounding of a value below the mathematical open boundary. Clean responses ranged from  $-2.757962$  to  $2.997449$ . Unnormalized and normalized weights are

$$\begin{aligned} \tilde{w}_{im} &= a_{im} q_{im} \exp(-\beta r_{im}), \\ w_{im} &= \frac{\tilde{w}_{im}}{\sum_k \tilde{w}_{ik}}. \end{aligned} \quad (3)$$

with  $\beta = 0.25$ . Fused logits are  $\ell_i = \sum_m w_{im} \mu_{im}$ . The implementation applies a  $10^{-6}$  lower guard to the normalization denominator, but the guard is inactive under the stated input constraints because every observation has at least one available group and  $q_{im} \geq 0.01$  after clipping.

This construction differs from an unconstrained softmax gate in three testable ways. First, the weight is factorized into availability, an optional externally measured quality term and one bounded operational response; it is not an unrestricted function of the concatenated sensor representation. Second, the perturbed group is known during paired degradation, so its response and normalized weight receive directional supervision rather than only a task-gradient signal. Third, the same response that controls fusion is regularized by a group-specific predictive loss. Thus the response is operationally coupled to group evidence, although it is not asserted to be a physical health variable.

The factorization also gives two elementary stability properties. For an available group, fixed  $q$  and any  $k \neq m$ ,

$$\begin{aligned} \frac{\partial w_{im}}{\partial r_{im}} &= -\beta w_{im}(1 - w_{im}) \leq 0, \\ \frac{\partial w_{ik}}{\partial r_{im}} &= \beta w_{ik} w_{im} \geq 0. \end{aligned} \tag{4}$$

For two available groups with equal quality, bounded responses imply

$$e^{-2\beta B} < \frac{\tilde{w}_{im}}{\tilde{w}_{ik}} < e^{2\beta B}. \tag{5}$$

At  $B = 3$  and  $\beta = 0.25$ , one response alone can therefore create at most a 4.48-fold unnormalized-weight ratio between equal-quality groups. These properties do not guarantee that the learned ordering is correct; the experiments separately test that question.

The response regularizer, retained as  $\mathcal{L}_{\text{score}}$  in the implementation for reproducibility, combines a bounded heteroscedastic-style term and a boundary penalty. For a minibatch  $\mathcal{B}$ , write  $s_{im} = \omega_{y_i} [-\log \text{softmax}(\mu_{im})_{y_i}]$ , where  $\omega_c = n/(Cn_c)$  is the training-set class weight, and let  $h_{im} = e^{-r_{im}} s_{im} + r_{im}$ . Then

$$\mathcal{L}_{\text{score}} = \frac{\sum_{i \in \mathcal{B}} \sum_m a_{im} h_{im}}{\sum_{i \in \mathcal{B}} \sum_m a_{im}}, \tag{6}$$

$$\mathcal{L}_{\text{bound}} = \frac{\sum_{i \in \mathcal{B}} \sum_m a_{im} (r_{im}/B)^4}{\sum_{i \in \mathcal{B}} \sum_m a_{im}}. \tag{7}$$

Because cross-entropy is non-negative and  $r_{im} > -B$ ,  $\mathcal{L}_{\text{score}} > -B$ . The complete objective therefore has a finite lower bound. This statement concerns optimization stability, not calibration of the bounded operational response.

### 3.3 Paired degradation and directional losses

For each minibatch observation, one available group  $m^*$  is drawn from a fixed training prior. A retention factor  $\rho_i \sim \text{Uniform}(0.15, 0.50)$  controls additive Gaussian degradation with scale  $2.5(1 - \rho_i)$ . In half of the pairs,  $q_{im^*}$  is multiplied by  $\rho_i$ ; otherwise it is unchanged. Let primed symbols denote the degraded view and  $d_i = 1 - \rho_i$ .

The clean and degraded task losses use the class-weighted mean implemented by PyTorch. With  $e_i(\ell) = -\log \text{softmax}(\ell)_{y_i}$ ,

$$\begin{aligned} \mathcal{L}_{\text{task}} &= \frac{\sum_{i \in \mathcal{B}} \omega_{y_i} e_i(\ell_i)}{\sum_{i \in \mathcal{B}} \omega_{y_i}}, \\ \mathcal{L}_{\text{deg}} &= \frac{\sum_{i \in \mathcal{B}} \omega_{y_i} e_i(\ell'_i)}{\sum_{i \in \mathcal{B}} \omega_{y_i}}. \end{aligned} \tag{8}$$

170 Let  $\mathcal{V}_{\mathcal{B}} = \{i \in \mathcal{B} : \sum_m a_{im} > 1\}$  denote observations for which the selected group can be re-  
 171 moved while another group remains. For  $i \in \mathcal{V}_{\mathcal{B}}$ , the removal prediction  $\mathbf{p}_i^{(-m^*)}$  defines a severity-  
 172 interpolated target  $\mathbf{t}_i$ ; write  $\hat{\mathbf{t}}_i = \text{stopgrad}(\mathbf{t}_i)$ :

$$\begin{aligned} \mathbf{t}_i &= (1 - d_i)\mathbf{p}_i + d_i\mathbf{p}_i^{(-m^*)}, \\ \mathcal{L}_{\text{cons}} &= \frac{1}{|\mathcal{V}_{\mathcal{B}}|} \sum_{i \in \mathcal{V}_{\mathcal{B}}} \|\mathbf{p}'_i - \hat{\mathbf{t}}_i\|_2^2. \end{aligned} \tag{9}$$

173 If  $\mathcal{V}_{\mathcal{B}}$  is empty,  $\mathcal{L}_{\text{cons}}$  and the removal-dependent directional terms are omitted, equivalently set  
 174 to zero. Directional terms operate on the affected normalized weight and raw response and are  
 175 averaged over eligible observations. With  $[z]_+ = \max(0, z)$  and  $\Delta w_i = w'_{im^*} - w_{im^*}$ ,

$$\mathcal{L}_{\text{mono}} = |\mathcal{V}_{\mathcal{B}}|^{-1} \sum_{i \in \mathcal{V}_{\mathcal{B}}} [\Delta w_i]_+. \tag{10}$$

176 With  $\gamma_i = 0.5d_i/0.85$  and  $\Delta r_i = \text{stopgrad}(r_{im^*}) - r'_{im^*}$ , the response-ranking term is

$$\mathcal{L}_{\text{rank}} = |\mathcal{V}_{\mathcal{B}}|^{-1} \sum_{i \in \mathcal{V}_{\mathcal{B}}} [\gamma_i + \Delta r_i]_+. \tag{11}$$

177 The final loss is

$$\begin{aligned} \mathcal{L} &= \mathcal{L}_{\text{task}} + 0.20\mathcal{L}_{\text{score}} \\ &\quad + 0.50\mathcal{L}_{\text{bound}} + 0.15\mathcal{L}_{\text{deg}} \\ &\quad + 0.10\mathcal{L}_{\text{cons}} + 0.05\mathcal{L}_{\text{mono}} \\ &\quad + 0.15\mathcal{L}_{\text{rank}}. \end{aligned} \tag{12}$$

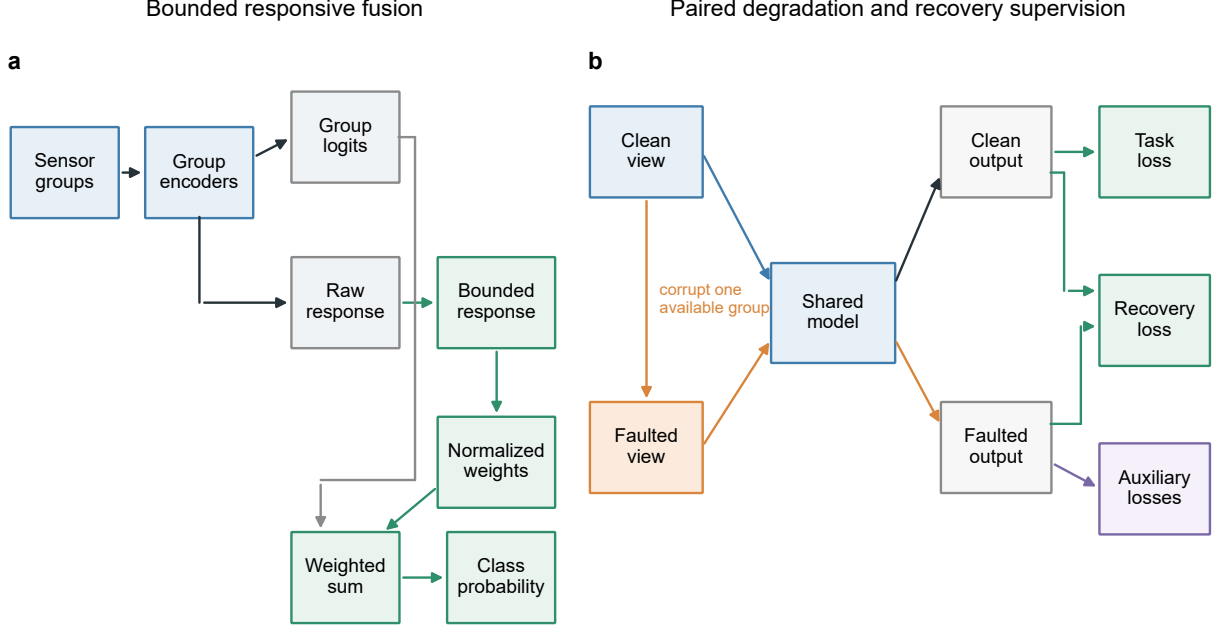


Figure 1: **Bounded responsive fusion and recovery supervision.** **a**, Each sensor group produces class logits and a bounded operational response. Availability, optional diagnostic quality and the response determine normalized fusion weights. Distinct connector ports separate the logit and weight contributions to the fused prediction. **b**, Clean and faulted views share model parameters. The detached clean prediction supervises recovery of the faulted prediction; proper-scoring and ranking terms are auxiliary controls.

### 3.4 Comparators and mechanism ablations

Early fusion concatenates zero-filled group features, masks and quality values before a  $128 \rightarrow 64 \rightarrow C$  multilayer perceptron; uniform fusion averages group logits. Both receive the same paired degradation and degraded-task loss as PDRF. CAGF concatenates four 32-dimensional group representations, masks and quality values before an unrestricted  $64 \rightarrow M$  softmax gate. It shares PDRF’s paired-degradation, consistency and normalized-weight monotonicity losses. Direct weight ranking adds a gate-weight margin, while the quality-only control uses  $a_{im}q_{im}$  without a learned response.

RO-AT-GATE tests stronger cross-group interaction. It adds learned group identities to 32-dimensional tokens and applies one four-head attention layer with a 64-unit feed-forward block; unavailable groups are masked. A shared  $32 + 2 \rightarrow 16 \rightarrow 1$  head maps each contextualized token, availability and quality to an unrestricted softmax gate. The 26,809-parameter model receives the same paired views and recovery objective as RO-CAGF. It is a sensor-group attention control, not a reproduction of a task-specific vision or language system [6, 12, 13].

RO-MER supplies a sample-dependent multi-expert control motivated by failure-resilient routing [14]. It combines one classifier per physical group with a joint 64-unit expert and a separate 64-unit router. The prediction uses the resulting  $(M + 1)$ -expert distribution. Losses requiring physical-group weights use the differentiable allocation

$$w_{im}^{\text{eff}} = \alpha_{im} + \alpha_{iJ} \frac{a_{im}}{\sum_k a_{ik}}. \quad (13)$$

The joint probability is distributed uniformly only for shared monotonicity and gate regularization; Eq. (13) does not alter the prediction. RO-MER has 35,811 parameters and uses the same faults, batches, seeds and checkpoint rule as RO-CAGF and RO-PDRF-Full.

The complete RO-MER objective is

$$\begin{aligned}\mathcal{L}_{\text{RO-MER}} = & \mathcal{L}_{\text{task}} + 0.15\mathcal{L}_{\text{deg}} + 0.10\mathcal{L}_{\text{cons}} + 0.05\mathcal{L}_{\text{mono}} \\ & + 0.30\mathcal{L}_{\text{rec}} + 0.05\mathcal{L}_{\text{Brier}} + 0.10\mathcal{L}_{\text{AUC}} + 0.005\mathcal{L}_{\text{router}},\end{aligned}\tag{14}$$

Here  $\mathcal{L}_{\text{router}}$  is the clean-plus-degraded Kullback–Leibler divergence from  $\mathbf{w}_i^{\text{eff}}$  to uniform allocation over available groups. RO-CAGF uses the same coefficients with native softmax weights; RO-PDRF-Full substitutes its response-interior penalty. Recovery supervision, proper scoring, pairwise ranking, paired views and task losses are otherwise matched.

We additionally adapt Quality-aware Multimodal Fusion (QMF) to the same four sensor groups while retaining its detached energy-confidence fusion and historical-loss ranking rule [18]. This QMF-PD control shares the present encoders, corruptions, optimizer, split and checkpoint criterion; it is a sensor-group adaptation rather than an exact reproduction of the original backbones or tasks. Implementation choices and sensitivity variants are specified in the Supplementary Information.

PDRF contains 19,740 trainable parameters on the chemical array, compared with 26,588 for CAGF and DWR, 35,811 for RO-MER, 26,182 for early fusion and 17,560 for uniform fusion. Ablations comprise paired degradation only (PD-ONLY), paired degradation plus normalized-weight monotonicity (PD-MONO), paired degradation plus response ranking (PD-RANK), the complete objective without  $\mathcal{L}_{\text{score}}$  (NO-LSCORE), and  $\mathcal{L}_{\text{score}}$  without  $\mathcal{L}_{\text{rank}}$  (LSCORE-NO-LRANK). A PDRF-NOQ model is trained and tested with all quality values fixed to one.

### 3.5 Recovery-oriented extension

Reducing an affected group’s weight did not ensure prediction recovery. RO-PDRF therefore retains the PDRF architecture and distills the degraded prediction from the detached clean-view prediction. Writing  $\hat{\mathbf{p}}_i = \text{stopgrad}(\mathbf{p}_i)$  gives

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{V}_{\mathcal{B}}|} \sum_{i \in \mathcal{V}_{\mathcal{B}}} D_{\text{KL}}[\hat{\mathbf{p}}_i \parallel \mathbf{p}'_i].\tag{15}$$

$\mathcal{L}_{\text{rec}}$  requires at least two available groups and is omitted when  $\mathcal{V}_{\mathcal{B}}$  is empty. This excluded  $0.386\% \pm 0.052\%$  of 5,933 training rows across seeds. PDRF plus  $0.30\mathcal{L}_{\text{rec}}$  defines RO-PDRF-Lite. Label-assisted, confidence-weighted, exponential-moving-average and agreement-gated teachers remain supplementary controls; the deployed selector uses no held-out label [58].

Second, multiclass Brier losses on clean and degraded predictions provide proper-scoring supervision,

$$\begin{aligned}\mathcal{L}_{\text{Brier}} = & \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\mathbf{p}_i - \mathbf{e}_{y_i}\|_2^2 \\ & + \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \|\mathbf{p}'_i - \mathbf{e}_{y_i}\|_2^2.\end{aligned}\tag{16}$$

The Brier term is unweighted; class weights remain confined to cross-entropy. A weak log barrier then discourages boundary occupancy without removing the quartic penalty. With  $\epsilon = 10^{-4}$ , define  $b_{\epsilon}(r) = -\log\{\max[\epsilon, 1 - (r/B)^2]\}$ ,  $\bar{b}_{im} = b_{\epsilon}(r_{im}) + b_{\epsilon}(r'_{im})$  and  $Z_{\mathcal{B}} = \sum_{i \in \mathcal{B}} \sum_m a_{im}$ :

$$\mathcal{L}_{\text{interior}} = Z_{\mathcal{B}}^{-1} \sum_{i \in \mathcal{B}} \sum_m a_{im} \bar{b}_{im}.\tag{17}$$



Unavailable groups are excluded by  $a_{im}$ ; the clamp is numerical. Finally, a smooth one-versus-rest pairwise loss supervises degraded-view class ordering. Let  $P_c$  and  $N_c$  denote positive and negative minibatch observations,  $\mathcal{C}_B = \{c : |P_c| > 0, |N_c| > 0\}$  and  $\phi_{ijc} = \text{softplus}[-(\ell'_{ic} - \ell'_{jc})]$ :

$$\begin{aligned}\mathcal{L}_{\text{AUC}} &= \frac{1}{|\mathcal{C}_B|} \sum_{c \in \mathcal{C}_B} A_c, \\ A_c &= \frac{\sum_{i \in P_c} \sum_{j \in N_c} \phi_{ijc}}{|P_c| |N_c|}.\end{aligned}\tag{18}$$

The pairwise term is omitted when no class has both outcomes [59]. RO-PDRF-Full adds  $0.30\mathcal{L}_{\text{rec}} + 0.05\mathcal{L}_{\text{Brier}} + 0.005\mathcal{L}_{\text{interior}} + 0.10\mathcal{L}_{\text{AUC}}$  to Eq. (12). RO-PDRF-Lite is the practical default, Full is the mechanistic architecture comparator and RO-PDRF-EMA is a Full-based calibration sensitivity. No test observation fits a model or ensemble parameter.

To separate recovery supervision from architecture, we also train recovery-oriented CAGF (RO-CAGF). It receives the same clean-view distillation, Brier and degraded-view ranking losses as RO-PDRF-Full. Its unrestricted softmax gate uses a weak KL-to-uniform regularizer (weight 0.005), whereas RO-PDRF-Full uses the response-interior penalty. The matched data, fixed faults, seeds, minibatches and checkpoint rules isolate architecture subject to this necessary regularizer difference. Regularizer grids, early/uniform-fusion controls, modality-dropout self-distillation and confidence-thresholded distillation are reported in the Supplementary Information [5, 6].

### 3.6 Cross-fitted selective recovery

Unconditional recovery can repair a wrong base prediction but can also overturn a correct one. We therefore treat routing as hard-decision risk control. Here “Lite” means that the router selects between PDRF and the RO-PDRF-Lite recovery endpoint; it does not merely denote a smaller selector. Both endpoints require one ordinary forward pass and have the same 19,740-parameter architecture. The heavier PDRF/RO-PDRF-Full pair is retained only as a supplementary mechanistic and legacy-routing sensitivity.

Batch 7 is divided chronologically into two disjoint halves. The first half fits one scalar temperature per endpoint by minimizing multiclass negative log-likelihood. The second contains two calibration-only realizations of each fault mechanism; only interventions that reach an available affected group enter selector development. Routing features use the temperature-scaled endpoint probabilities. The selected endpoint vector is emitted without probability mixing. Uncalibrated endpoint features, Platt scaling and isotonic calibration of the preference score are separate controls rather than replacements selected from test performance.

Lite-CF uses six prespecified inputs: base confidence, recovery confidence, their difference, base normalized entropy, recovery normalized entropy and their difference. It receives neither the class label nor fault type at inference. Let  $z_{ij}$  denote standardized feature  $j$ . Feature means and standard deviations are fitted only on the informative training rows inside each cross-fitting fold, then refitted on all informative calibration rows for the final model. An  $L_2$ -regularized logistic model produces the *conditional preference score*

$$\pi_i = \sigma \left( \gamma_0 + \sum_{j=1}^6 \gamma_j z_{ij} \right),\tag{19}$$

using calibration rows on which PDRF and RO-PDRF-Lite differ in correctness. The target equals one when recovery is correct and the base endpoint is wrong. Therefore  $\pi_i$  estimates a conditional preference on the label-selected subset  $b_i \neq r_i$ ; it does not estimate  $\Pr(r_i = 1, b_i = 0 \mid \mathbf{x}_i)$  or a

general reliability probability for all deployment rows. Lite-CF uses  $C = 0.05$ , the `lbfgs` solver, balanced class weights, tolerance  $10^{-4}$ , 2,000 maximum iterations and random state 0. We retain this model as the frozen historical router and explicitly compare it with all-row alternatives.

Coefficient fitting is separated from threshold selection by five-fold stratified group cross-fitting. Realizations and mechanisms from the same calibration observation remain in one fold. Candidate thresholds 0.50, 0.525,  $\dots$ , 0.90 are evaluated only on pooled out-of-fold scores. Selection minimizes realized harms, then maximizes selected accuracy and retained corrections; remaining ties use the highest threshold. The formal fits had a mean of 1.5 tied thresholds (range 1–4). Final coefficients are refitted on all informative rows while the threshold remains frozen.

For prospective deployment, mathematical fit is necessary but insufficient. We define an operational support gate requiring at least 25 events in each preference class, median out-of-fold AUROC at least 0.65 across 10 repeated grouped splits, no repeated AUROC below 0.60 and a threshold range no wider than 0.10. Both preference classes must also occur in every fitting fold. The gate was specified after the frozen experiment, so the 94.7%/9.7% estimate remains labelled ungated. Nevertheless, the gate-enforced policy is the primary deployment interpretation: a site that fails any condition must emit PDRF while collecting new calibration evidence. The released routing API returns probabilities by default and can optionally return selected endpoint identity and the conditional preference score; routed confidence remains endpoint-specific rather than globally calibrated.

We address the selected-sample problem with an all-row outcome  $o_i \in \{0, 1, 2\}$  denoting base better, equivalent hard-decision correctness or recovery better. A multinomial logistic model produces probabilities  $\eta_{ik} = \Pr(o_i = k \mid \mathbf{x}_i)$  and score  $d_i = \eta_{i2} - \eta_{i0}$ . A two-stage alternative first estimates  $\Pr(b_i \neq r_i \mid \mathbf{x}_i)$  on all rows and then multiplies it by the conditional base/recovery preference. Both stages are cross-fitted by original observation. These models target the unconditional decision problem without inverse-probability correction because all calibration rows and their three outcomes are observed.

Controls use the same two endpoint passes and calibration budget. They include random ranking, low base confidence, recovery-minus-base confidence, base-minus-recovery entropy, recovery-minus-base top-two logit margin, a depth-two tree with minimum leaf size 35, rank-normalized logistic features and class-conditional endpoint-prediction indicators. Complete operating curves rank rows within each fixed mechanism–fitted-pair cell. Curves are descriptive test audits; no point on a test curve selects the submitted operating policy. For matched controls, thresholds are selected only on calibration to match the frozen router’s calibration prevention or retention and are then applied unchanged to test rows.

We further probe sparse estimation through ten repeated grouped splits for each endpoint pair, standardized coefficient/sign distributions,  $C \in \{0.01, 0.05, 0.1, 0.5, 1, 5\}$ , calibration-size subsampling and five calibration-family mixtures (uniform and one 70%-dominant mechanism). Leave-one-mechanism-out and Gaussian-only calibration test mechanism transport. Severity transport fits selectors at scales 1, 2 or 3 and evaluates every scale without endpoint refitting; affected-group transport calibrates on group 1 and evaluates groups 1–4 at scale 3.

The deployment audit reuses the same endpoints, temperatures, selector and test realization for every stream. It includes the strict fault-applied-and-available subset, a complete 40%-prevalence mixed stream and a no-imposed-fault stream with the same fixed 20% group-missingness process. Probability quality is reported separately for endpoint-class agreement and disagreement. The primary outcomes are negative-transfer prevention, recovery retention and

$$U = V_C n_C - C_H n_H - \lambda(P - 1), \quad (20)$$

where  $n_C$  and  $n_H$  are retained corrections and realized harms per 10,000 observations,  $V_C$  is correc-

tion value,  $C_H$  is harm cost,  $P$  is mean forward-pass count and  $\lambda$  is a pass-cost proxy in correction-equivalents per 10,000 observations. We vary  $C_H/V_C$  over 1, 2, 5 and 10, imposed-fault prevalence and  $\lambda$ . Energy was not measured; forward passes and counted linear-layer FLOPs remain computational proxies.

Two threshold regimes are kept distinct. The safety threshold uses the frozen lexicographic rule: minimize calibration harms, then maximize selected accuracy and retained corrections. The application threshold maximizes  $n_C - (C_H/V_C)n_H$  on pooled out-of-fold calibration predictions for each declared cost ratio. Both are evaluated once on test. Always-base and always-recovery remain explicit endpoints. Reporting includes extra forward passes per retained and net correction, selected observations per net correction, model storage, router latency and orchestration residual beyond the sum of endpoint timings.

### 3.7 Chemical-sensor dataset and fixed faults

The UCI Gas Sensor Array Drift Dataset contains 13,910 observations from 16 metal-oxide sensors, six gas classes and 10 acquisition batches over 36 months [60, 61]. Each sensor contributes eight extracted response features. The principal grouping uses four consecutive sensors per group. All models receive the same grouping. Sensitivity analyses additionally use interleaved and five fixed random partitions; no partition is called a physical modality.

Batches 1–6 train the models. Their feature means and standard deviations are the only quantities used for standardization. The same transform is applied to batches 7–10, after which each feature is clipped to  $[-8, 8]$  before sensor grouping. Thus every reported “training range” is the empirical minimum–maximum after training-only standardization and clipping; it is not a physical sensor limit. The first 60% of Batch 7 selects checkpoints, and the remaining 40% is reserved for calibration; the routing experiment further divides that calibration partition into temperature and selector halves. Batches 8–10 are never used to fit preprocessing, endpoints, temperatures, selectors or thresholds.

Controlled faults are imposed after this training-only preprocessing in grouped standardized space. The formal test assigns 40% of observations a scale-3 intervention on group 1 and fixes  $q = 1$ . Fault assignment and 20% group missingness are generated independently to isolate their factorial effects. This is a stress design, not a claim that all missingness–corruption combinations occur naturally. Of 4,364 test observations, 1,765 are assigned the intervention; group 1 remains available and receives it in 1,436, whereas it is masked in 329. The strict recovery estimand is the former *fault-applied-and-available* subset. Fault indices, missingness masks and intervention values are fixed across methods and optimization seeds.

The complete test set contains 294, 470 and 3,600 observations in batches 8, 9 and 10. Its six class counts are 691, 685, 740, 708, 821 and 719. The corresponding strict-subset batch counts are 91, 165 and 1,180, and its Gaussian class counts are 230, 220, 230, 253, 266 and 237. Complete mechanism-by-batch class tables are supplied in the Supplementary Information.

Checkpoint selection and temperature calibration use separate fault-free views with independently generated 10% group missingness; no controlled Gaussian, offset, drift or stuck-at intervention is added to either subset. The checkpoint criterion is unweighted cross-entropy on the complete selection subset. For the cross-fitted deployment run, temperature is fitted on the first chronological half of the calibration partition, not an affected subset, by minimizing unweighted multiclass negative log-likelihood. The resulting scalar is specific to each fitted seed and method and is applied unchanged to every deployment stream. Controlled-fault and no-fault deployment streams use the same fixed 20% group-missingness process so that fault prevalence is the changing factor. Earlier natural-view analyses without imposed test missingness remain supplementary

sensitivity evidence.

The expanded fault study crosses four prespecified mechanisms with realizations 70001–70008 and fitted seeds 101–105. Model fitting always uses the Gaussian paired degradation defined above; offset, drift and stuck-at are test-time interventions only. Gaussian faults add independent standardized noise. Offset faults add a featurewise signed displacement. Drift multiplies that displacement by a monotone 0.25–1.00 trajectory over held-out acquisition order. Stuck-at faults replace the affected group’s standardized features by a fixed signed vector. Within each realization, all mechanisms share affected rows and missingness masks, while intervention values remain fixed across methods. These are controlled repeats on one instrument.

Table 2 translates the formal scale into empirical training-distribution displacement. At scale 3, median absolute shifts were 1.89–3.10 training standard deviations and median percentile shifts were 0.36–0.56. Only 9.4–45.7% of corrupted values remained inside the training 1st–99th percentile interval. The formal setting is therefore a severe standardized stress test, not a typical natural-failure model. Scale-1 and scale-5 summaries and representative inverse-transformed feature examples are reported in the Supplementary Information.

Table 2: Fault-plausibility audit in the strict subset. Fractions are calculated featurewise against batches 1–6. “Training range” denotes the observed training minimum–maximum, not a physical sensor limit.

| Mechanism (scale 3) | Median $ \Delta z $ | Median percentile shift | Within training 1–99% | Within training range |
|---------------------|---------------------|-------------------------|-----------------------|-----------------------|
| Gaussian            | 2.015               | 0.360                   | 0.379                 | 0.609                 |
| Offset              | 3.000               | 0.538                   | 0.332                 | 0.632                 |
| Drift               | 1.888               | 0.440                   | 0.457                 | 0.697                 |
| Stuck-at            | 3.105               | 0.557                   | 0.094                 | 0.656                 |

Exploratory controls vary diagnostic-quality metadata, leave-one-group-out training, response bounds, weighting slope, ranking weight, fault group, severity and prevalence. They share observations and fitted models, are not treated as independent replications and do not replace the prespecified main setting. Full specifications and results are reported in the Supplementary Information.

### 3.8 Independent hydraulic single-rig sensitivity analysis

The UCI Condition Monitoring of Hydraulic Systems dataset contains 2,205 raw 60-s cycles from an experimental hydraulic rig and native condition labels for cooler, valve, pump and accumulator states (DOI: 10.24432/C5CW21) [22, 62]. We retain 1,449 cycles marked as stable. Fourteen physical sensors form five groups by measured quantity: six pressure sensors, two flow sensors, four temperature sensors, motor power and vibration. This grouping is defined before model fitting and follows the measurement system rather than a random partition.

Each sensor is summarized by distributional and trend statistics. Unequal group lengths are aligned to 48 inputs; a zero-padded native-order representation audits possible smoothing. The main split preserves acquisition order within every native condition class, assigns contiguous 60%/15%/10%/15% blocks and removes five cycles at each boundary. A global chronological split would omit some condition classes from training. After guard-band removal, the test counts are 72/72/74 for the three cooler states, 54/54/54/56 for the four valve states, 74/72/72 for the three pump states and 56/54/54/54 for the four accumulator states. Recovery-objective-matched RO-PDRF-Full and RO-CAGF use five matched seeds; controlled pressure- and vibration-group faults affect the same 40% of test cycles. Component-condition labels are native to the rig, whereas sen-

393 sor faults remain interventions. Representation and random-split sensitivities are supplementary  
394 [23–25].

### 395 **3.9 Real-operational AHU direct-diagnosis boundary analysis**

396 We additionally use the open real-operational AHU dataset from an auditorium, a hospital and  
397 a large office (Figshare DOI: 10.6084/m9.figshare.27147678.v3) [26, 27]. The binary task uses  
398 native, expert-annotated temperature-sensor-fault labels and no artificial test corruption. Seven  
399 measurements form five physical groups; each building uses a chronological 60%/15%/10%/15%  
400 split and five matched seeds. The untouched test periods contain 228/9,354 fault records in the  
401 auditorium (2.44%), 1,495/9,362 in the hospital (15.97%) and 346/19,115 in the office (1.81%).  
402 Return-air/supply-air subtype counts are 179/49, 1,487/8 and 293/53, respectively. Because the  
403 label is the sensor fault itself and no independent downstream state is available, this is a negative  
404 task-boundary audit, not downstream recovery validation. Detailed shift, adaptation, risk-coverage  
405 and conformal analyses are reported in the Supplementary Information [41–44].

### 406 **3.10 Training, metrics and statistical analysis**

407 AdamW uses learning rate  $2 \times 10^{-3}$ , weight decay  $10^{-4}$ , batch size 256 and gradient clipping at  
408 5. Chemical-array training uses at most 60 epochs, hydraulic training uses at most 80 and AHU  
409 training uses at most 12. Checkpoints minimize unweighted selection cross-entropy and stop after  
410 nine epochs without an improvement greater than  $10^{-4}$ . RO-PDRF-Lite, RO-PDRF-Full, RO-  
411 PDRF-EMA and all teacher controls use this identical criterion; no test metric selects a checkpoint  
412 or variant. Class weights are  $n/(Cn_c)$ . No dropout or batch normalization is used.

413 The primary deployment outcomes are negative-transfer prevention, recovery retention and the  
414 utility in Eq. (20). Macro-AUROC, macro-AUPRC, NLL, multiclass Brier score and calibration  
415 are secondary. Equal-width ECE partitions maximum class probability into 15 intervals on  $[0, 1]$ .  
416 Equal-mass ECE splits the sorted maximum probabilities into 15 groups. Classwise ECE applies 15  
417 equal-width bins to each one-versus-rest probability and averages the six values. Classwise AUROC,  
418 AUPRC and recall are archived. Temperature scaling is fitted separately for each endpoint and  
419 optimization seed on the disjoint first half of Batch 7.

420 The deployment analyses are frozen as CEE-CF10-R2 for the Full endpoint and CEE-CF10-  
421 R2-LITE for the primary Lite endpoint. Within each seed, one endpoint pair is reused across all  
422 four faults and selector comparisons. Test realization 70001 is fixed. Means over the strict set  
423 give equal weight to the four mechanism-seed cells. Clean/no-fault summaries contain one value  
424 per fitted pair. Mechanism-specific effects use paired Student- $t$  intervals across the 10 fitted model  
425 pairs while holding the mechanism fixed. They are labelled fitted-pair or fitting-variation intervals;  
426 all source files also report every point, median, interquartile range and full range. No interval is  
427 interpreted as a significance test.

428 The strict subset contains 91 Batch 8, 165 Batch 9 and 1,180 Batch 10 observations per  
429 mechanism, so Batch 10 supplies 82.2% of the observation-weighted result. The complete  
430 mechanism $\times$ batch $\times$ method matrix reports accuracy, macro-AUROC, correction and harm op-  
431 portunities, prevention, retention and utility. We separately report pooled observation-weighted  
432 performance, an equal-batch average, the worst acquisition batch and pooled Batches 8–9 versus  
433 Batch 10. Batch-aware intervals hold mechanism, batch and test rows fixed and vary only the 10  
434 fitted endpoint pairs.

435 The previous four-mechanism crossed bootstrap is retained only as a labelled sensitivity in  
436 the Supplementary Information. It is not used for new headline inference because four top-level

mechanism clusters do not support a reliable mechanism-population bootstrap. Exact Wilcoxon tests across optimization seeds are likewise reported only as fitting-stability diagnostics. Hydraulic summaries use a task–fault–seed hierarchy; AHU results remain building-specific. The strict estimand and selector analyses followed reviewer-driven audit and are not described as preregistered confirmatory analyses.

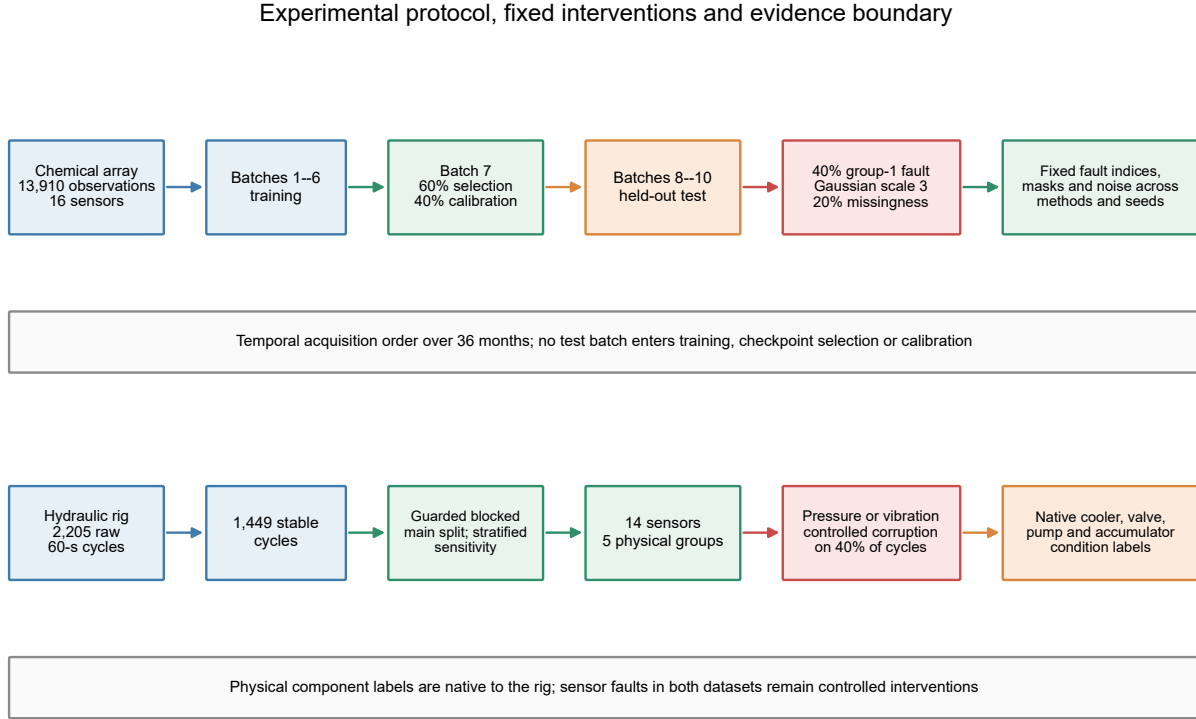


Figure 2: **Integrated experimental protocol and controlled fault design.** The upper lane follows the chemical-array acquisition order from batches 1–6 training through Batch 7 checkpoint selection/calibration to held-out batches 8–10; fault indices, masks and intervention values are fixed across methods and seeds. The lower lane follows the independent hydraulic rig from stable 60-s cycles to the guarded within-condition blocked main analysis; the earlier stratified split is retained only as a sensitivity analysis. Component-condition labels are native to the rig, whereas chemical, pressure and vibration sensor corruptions remain controlled interventions.

## 4 Results

### 4.1 Always-base context and the support-gated deployment policy

The strict subset contains 1,436 observations per mechanism on which the assigned intervention reached an available group. Across the 40 fixed mechanism–fitted-pair cells, PDRF achieved accuracy 0.4869 and macro-AUROC 0.8115. Unconditional RO-PDRF-Lite increased these values to 0.5002 and 0.8175, but it realizes every possible correction and every possible harm. Always PDRF is therefore the maximum-safety reference: 100% negative-transfer prevention, 0% recovery retention and zero routing utility by definition (Table 3).

The historical frozen Lite-CF result is explicitly ungated. It selected recovery on 52.2% of strict rows, prevented 94.7% of recovery-induced negative transfers and retained 9.7% of available corrections, yielding accuracy 0.4885, macro-AUROC 0.8151 and equal-cost utility +15.8 per 10,000. Only 5/10 fitted endpoint pairs passed the prespecified event, discrimination and threshold-stability support criteria. Under the primary prospective policy, unsupported pairs revert to PDRF; this reduced recovery selection to 31.1%, increased prevention to 96.2% and reduced retention to 6.6%. Its macro-AUROC was 0.8132 and equal-cost utility was +7.5. The support-gated policy is thus safer than the frozen router but only 0.0017 AUROC above always PDRF on this fixed test set.

Table 3: **Strict-subset endpoints and calibration-frozen policies.** Values are means over 40 fixed mechanism–fitted-pair cells. Prevention and retention have different opportunity denominators.  $U/10^4$  uses equal correction and harm values and no compute penalty. The support gate is the primary prospective policy; the 94.7%/9.7% row is the ungated frozen result. All individual points and fitted-pair summaries are supplied as source data.

| Policy                            | Accuracy | Macro-AUROC | Prevent (%) | Retain (%) | Select (%) | $U/10^4$ |
|-----------------------------------|----------|-------------|-------------|------------|------------|----------|
| Always PDRF                       | 0.4869   | 0.8115      | 100.0       | 0.0        | 0.0        | 0.0      |
| Always RO-PDRF-Lite               | 0.5002   | 0.8175      | 0.0         | 100.0      | 100.0      | 132.1    |
| Frozen Lite-CF (ungated)          | 0.4885   | 0.8151      | 94.7        | 9.7        | 52.2       | 15.8     |
| Prospective support gate          | 0.4877   | 0.8132      | 96.2        | 6.6        | 31.1       | 7.5      |
| All-row multinomial (safety)      | 0.4879   | 0.8118      | 97.2        | 7.3        | 3.0        | 9.6      |
| All-row multinomial (utility 1:1) | 0.4977   | 0.8160      | 37.5        | 70.3       | 66.8       | 107.9    |

The 52.2% selection rate and 9.7% retention rate have different denominators. Selection is normalized by all 1,436 strict observations, whereas retention is normalized only by  $N_C$ , the sparse rows on which recovery alone corrects PDRF. Selection can therefore be high because the router also chooses recovery on endpoint-class agreements and rows where both endpoints share correctness status. Table 4 reports the absolute event counts rather than only opportunity-normalized rates.

Table 4: **Absolute strict-subset opportunities and routed events by mechanism.**  $N_C$  and  $N_H$  are correction and harm opportunities;  $n_C$  and  $n_H$  are retained corrections and realized harms. Counts are mean absolute events per 1,436-observation fitted pair, with ranges across 10 fitted pairs in parentheses. Select uses all observations; prevent and retain use  $N_H$  and  $N_C$ , respectively.

| Mechanism | $N_C$        | $N_H$        | $n_C$      | $n_H$      | Select (%) | Prevent (%) | Retain (%) |
|-----------|--------------|--------------|------------|------------|------------|-------------|------------|
| Gaussian  | 45.9 (36–70) | 18.9 (13–32) | 6.2 (0–16) | 1.5 (0–4)  | 51.9       | 91.2        | 13.5       |
| Offset    | 33.6 (10–49) | 16.9 (3–37)  | 2.7 (0–9)  | 1.9 (0–15) | 54.1       | 93.6        | 8.1        |
| Drift     | 24.7 (5–43)  | 13.6 (4–29)  | 1.9 (0–7)  | 0.7 (0–6)  | 53.9       | 97.3        | 11.1       |
| Stuck-at  | 38.7 (16–86) | 17.6 (2–36)  | 3.1 (0–15) | 0.7 (0–4)  | 48.7       | 96.6        | 6.1        |

The routed probability vector improved on PDRF for NLL, Brier score, ECE and macro-AUPRC, but remained below unconditional RO-PDRF-Lite for every strict-set probability metric. On the complete 40%-prevalence stream, Lite-CF/PDRF/RO-PDRF-Lite macro-AUROC values were 0.8441/0.8426/0.8451, and NLL values were 2.1386/2.2131/2.1135. On the no-imposed-fault stream, their macro-AUROC values were 0.8629/0.8631/0.8623 and NLL values were 1.8636/1.9002/1.8626. Lite-CF prevented 94.2% of no-fault negative transfers but retained only 1.7% of no-fault corrections, giving −5.3 net correct decisions per 10,000. A low residual-harm rate—the observation-level form of a harmful switch—therefore does not imply that the routed probability vector is uniformly better.

## 4.2 All-row routing and simple controls exposed the selected-sample limit

Each calibration fit contained a mean of 36.1 base-better, 1,771.2 equivalent and 53.6 recovery-better rows. The all-row multinomial model used all three outcomes rather than fitting only the 89.7 correctness disagreements. Its pooled out-of-fold confusion matrix classified 73.7% of base-better, 81.8% of equivalent and 56.3% of recovery-better rows correctly; mean out-of-fold accuracy, NLL and three-class Brier score were 0.809, 0.523 and 0.260. The model therefore reduced the estimand mismatch but did not make the rare endpoint-preference classes easy to identify.

At its safety threshold, the all-row router achieved 97.2% prevention and 7.3% retention with only 3.0% recovery selection and +9.6 equal-cost utility. Complete fixed-test curves additionally compare the conditional model, two-stage model, random ranking, confidence difference, entropy difference, base uncertainty, logit-margin difference and shallow tree (Fig. 3a). No comparator curve empirically dominated the frozen operating point in both mean prevention and mean retention. Thresholds matched on calibration did not remain exactly matched on test: the random control at the calibration-prevention target produced 97.8% test prevention and 2.1% retention, whereas its calibration-retention target produced 79.4% prevention and 20.5% retention. The entropy rule gave 97.4%/7.5% and 85.3%/21.8% in the corresponding regimes. This transport error is why no simple test-curve point replaces the submitted policy.

Class-conditional logistic indicators improved disagreement preference AUROC from 0.562 to 0.680 and produced +85.0 equal-cost utility, but prevention fell to 71.5%. A direct logit-margin utility rule produced +118.4 utility with only 19.2% prevention. Rank-normalized features retained 94.3% prevention but only 8.0% of corrections. These controls show that greater average utility is achievable by accepting more harmful transfers; they do not dominate the safety policy.

Mechanism-fixed paired intervals replace the four-cluster bootstrap (Table 5). On the present fixed test set, Gaussian accuracy, AUROC and equal-cost utility differences were positive across fitted model pairs. Stuck-at AUROC was also positive, whereas its accuracy and utility intervals crossed zero. Offset and drift intervals crossed zero for most hard-decision quantities. Gaussian AUROC differences were positive within batches 8, 9 and 10 (+0.0062, +0.0077 and +0.0061). These intervals quantify fitting variation on fixed observations, not a population of devices or fault mechanisms.

Table 5: **Mechanism-fixed paired differences for the hard-decision objective.** Values are Lite-CF minus PDRF. Brackets are paired 95% Student- $t$  intervals across 10 fitted model pairs while holding mechanism and test observations fixed. Equal-cost utility uses  $V_C = C_H = 1$  and  $\lambda = 0$ . These are conditional fitting-stability summaries, not population intervals.

| Fixed mechanism | $\Delta$ accuracy          | $\Delta$ macro-AUROC       | Equal-cost $U/10^4$    |
|-----------------|----------------------------|----------------------------|------------------------|
| Gaussian        | +0.0033 [+0.0005, +0.0061] | +0.0064 [+0.0034, +0.0093] | +32.73 [+4.94, +60.52] |
| Offset          | +0.0006 [−0.0012, +0.0024] | +0.0028 [0.0000, +0.0057]  | +5.57 [−12.44, +23.58] |
| Drift           | +0.0008 [−0.0006, +0.0022] | +0.0018 [−0.0002, +0.0037] | +8.36 [−5.69, +22.41]  |
| Stuck-at        | +0.0017 [−0.0004, +0.0037] | +0.0034 [+0.0005, +0.0063] | +16.71 [−3.92, +37.35] |

The formal scale 3 intervention is a severe stress test, so we repeated the frozen test protocol at scale 1 without refitting endpoints, temperatures, selector coefficients or threshold. Lite-CF then reached 0.8357 macro-AUROC versus 0.8344 for PDRF, prevented 96.0% of negative transfers while retaining 5.3% of corrections and produced equal-cost utility +4.00 per 10,000 strict observations (Table 6). Milder corruption improved absolute endpoint performance but left fewer useful recovery opportunities, so safety prevention did not imply greater practical benefit.



Table 6: **Mild and formal controlled-corruption severity.** Values are equal-weight means over four fixed mechanisms and 10 fitted model pairs on the strict subset. The scale 1 audit reuses the scale 3-calibrated selector without refitting or retuning.  $U/10^4$  uses equal correction and harm values with no pass penalty.

| Scale | PDRF acc. | Lite-CF acc. | PDRF AUROC | Lite-CF AUROC | Prevent (%) | Retain (%) | $U/10^4$ |
|-------|-----------|--------------|------------|---------------|-------------|------------|----------|
| 1     | 0.5301    | 0.5305       | 0.8344     | 0.8357        | 96.0        | 5.3        | +4.00    |
| 3     | 0.4869    | 0.4885       | 0.8115     | 0.8151        | 94.7        | 9.7        | +15.84   |

### 4.3 Utility-aligned thresholds changed which policy was preferred

The frozen safety threshold and application-specific utility thresholds answer different questions. At equal correction and harm values, the frozen router yielded +15.8 and the support gate +7.5 net corrections per 10,000, whereas utility-selected all-row and two-stage routers yielded +107.9 and +110.6. Always recovery still gave the largest utility, +132.1, at the one-to-one ratio. When one harm cost two corrections, always recovery fell to +15.5, while calibration-selected all-row and two-stage routing yielded +30.8 and +37.8. Thus selective routing added decision value at the intermediate cost ratio even though the lexicographic safety threshold did not.

This advantage did not extend to high harm costs. At ratios 5 and 10, every learned or recovery policy had negative test utility and always PDRF, with utility zero, dominated (Table 7; Fig. 3c). The all-row safety threshold was the least negative learned policy at ratio 5 (−8.5), but it still did not justify routing. Every application threshold was selected on pooled out-of-fold calibration predictions before test evaluation. The failure at high ratios is therefore transport evidence, not a test-tuned threshold choice. Complete-stream utility also remained negative at 0% and 10% controlled-fault prevalence. Deployment must fix expected prevalence,  $V_C$ ,  $C_H$  and compute cost before choosing a threshold.

Table 7: **Fixed-safety and utility-selected policies.** Entries are net decision utility per 10,000 strict observations, with  $V_C = 1$ ,  $\lambda = 0$  and the indicated harm ratio. Utility thresholds are selected only on out-of-fold calibration predictions; endpoint and safety policies retain their frozen decisions. Always PDRF has utility zero by definition.

| $C_H/V_C$ | Always base | Always recovery | Frozen safety | Support gate | All-row safety | All-row utility | Two-stage utility |
|-----------|-------------|-----------------|---------------|--------------|----------------|-----------------|-------------------|
| 1         | 0.0         | 132.1           | 15.8          | 7.5          | 9.6            | 107.9           | 110.6             |
| 2         | 0.0         | 15.5            | 7.5           | 0.7          | 5.0            | 30.8            | 37.8              |
| 5         | 0.0         | −334.4          | −17.6         | −19.7        | −8.5           | −20.0           | −33.6             |
| 10        | 0.0         | −917.7          | −59.4         | −53.6        | −31.2          | −95.9           | −86.0             |

### 4.4 Severity, affected-group, mechanism and batch transport

Selector transport was evaluated without changing endpoint architecture, endpoint training data or the fixed test realization. A scale-3 conditional selector applied at test scales 1, 2 and 3 yielded equal-cost utilities +4.0, +12.5 and +15.8, respectively. Conversely, a scale-1 selector applied at scale 3 yielded +5.9. Across the full  $3 \times 3$  calibration-to-test matrix, conditional-router utility ranged from +2.1 to +17.9 per 10,000 (Fig. 4a). Positive means therefore did not imply severity-invariant thresholds or event rates.

Calibration on affected group 1 also transferred unevenly. Conditional Lite-CF prevention/retention changed from 94.7%/9.7% on group 1 to 80.8%/17.8%, 93.0%/21.1% and 90.2%/17.2% on groups 2–4. Equal-cost utility was +15.8, +6.1, +60.3 and +31.0, but the

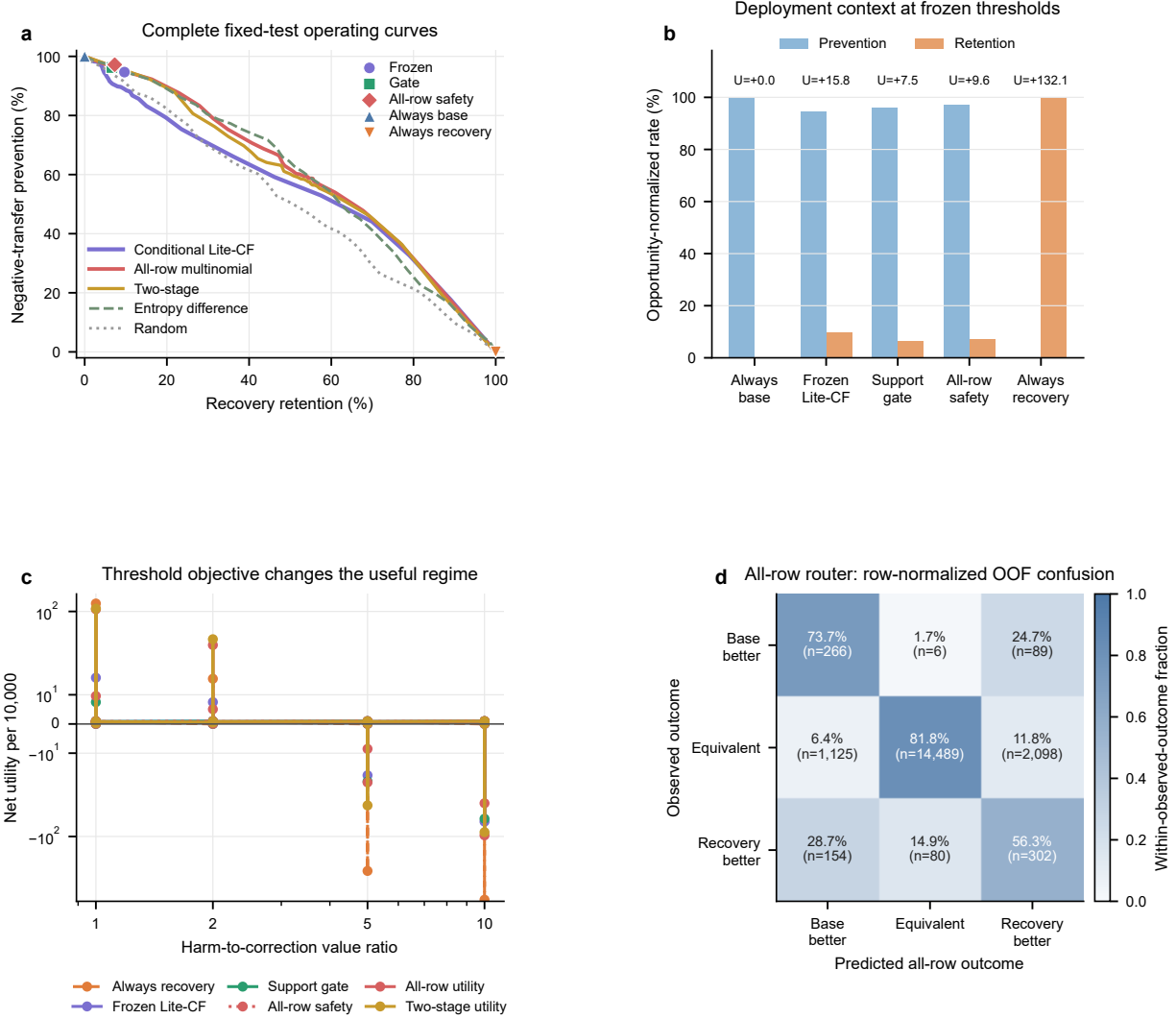
group-2 fitted-pair interval was  $-16.3$  to  $+28.4$ . The threshold is thus not transferable merely because sensor groups share a representation size.

Leave-one-mechanism-out conditional calibration produced mean utilities  $+27.9$ ,  $+10.4$ ,  $+1.4$  and  $+15.3$  on Gaussian, offset, drift and stuck-at tests, respectively. Every fitted-pair interval included zero; the drift range was  $-41.8$  to  $+20.9$ . Gaussian-only calibration and 70%-dominant calibration mixtures likewise changed mechanism-specific prevention, retention and utility. These audits support the prospective support gate rather than a universal selector.

Batch composition materially changed the aggregate. The 91/165/1,180 strict observations in Batches 8/9/10 make the observation-weighted result predominantly a Batch 10 result. Frozen Lite-CF macro-AUROC was 0.8151 overall, 0.8574 under equal batch weight, 0.8022 in the worst batch, 0.8747 in pooled Batches 8–9 and 0.8022 in Batch 10 (Table 8). Support-gated equal-cost utility was  $+7.5$  overall but  $-2.5$  in the worst batch. The complete mechanism $\times$ batch $\times$ method file reports opportunities and all seven requested outcomes for each fitted pair.

Table 8: **Batch-aware strict-set macro-AUROC.** Overall values pool observations within each fixed mechanism–fitted-pair cell; equal-batch values give Batches 8, 9 and 10 equal weight; worst batch is the minimum batch mean. Batches 8–9 are pooled and compared with Batch 10. These are summaries of one fixed instrument, not temporal-population intervals.

| Policy                   | Overall | Equal-batch | Worst batch | Batches 8–9 | Batch 10 |
|--------------------------|---------|-------------|-------------|-------------|----------|
| Always PDRF              | 0.8115  | 0.8528      | 0.7990      | 0.8689      | 0.7990   |
| Always RO-PDRF-Lite      | 0.8175  | 0.8607      | 0.8044      | 0.8780      | 0.8044   |
| Frozen Lite-CF           | 0.8151  | 0.8574      | 0.8022      | 0.8747      | 0.8022   |
| Prospective support gate | 0.8132  | 0.8550      | 0.8005      | 0.8721      | 0.8005   |
| All-row utility 1:1      | 0.8160  | 0.8591      | 0.8029      | 0.8771      | 0.8029   |



**Figure 3: Selective recovery as an explicit decision policy.** **a**, complete fixed-test prevention–retention curves averaged over 40 mechanism–fitted-pair cells. Curves are descriptive; marked policies use calibration-frozen thresholds. **b**, always-base, always-recovery, frozen conditional, support-gated and all-row safety policies. Utilities use equal correction and harm values. **c**, net utility under fixed-safety and calibration-selected utility thresholds. The symmetric-log axis preserves small values while showing the failure at high harm cost. **d**, row-normalized out-of-fold confusion matrix for the all-row multinomial outcome. Counts pool the 10 calibration fits.

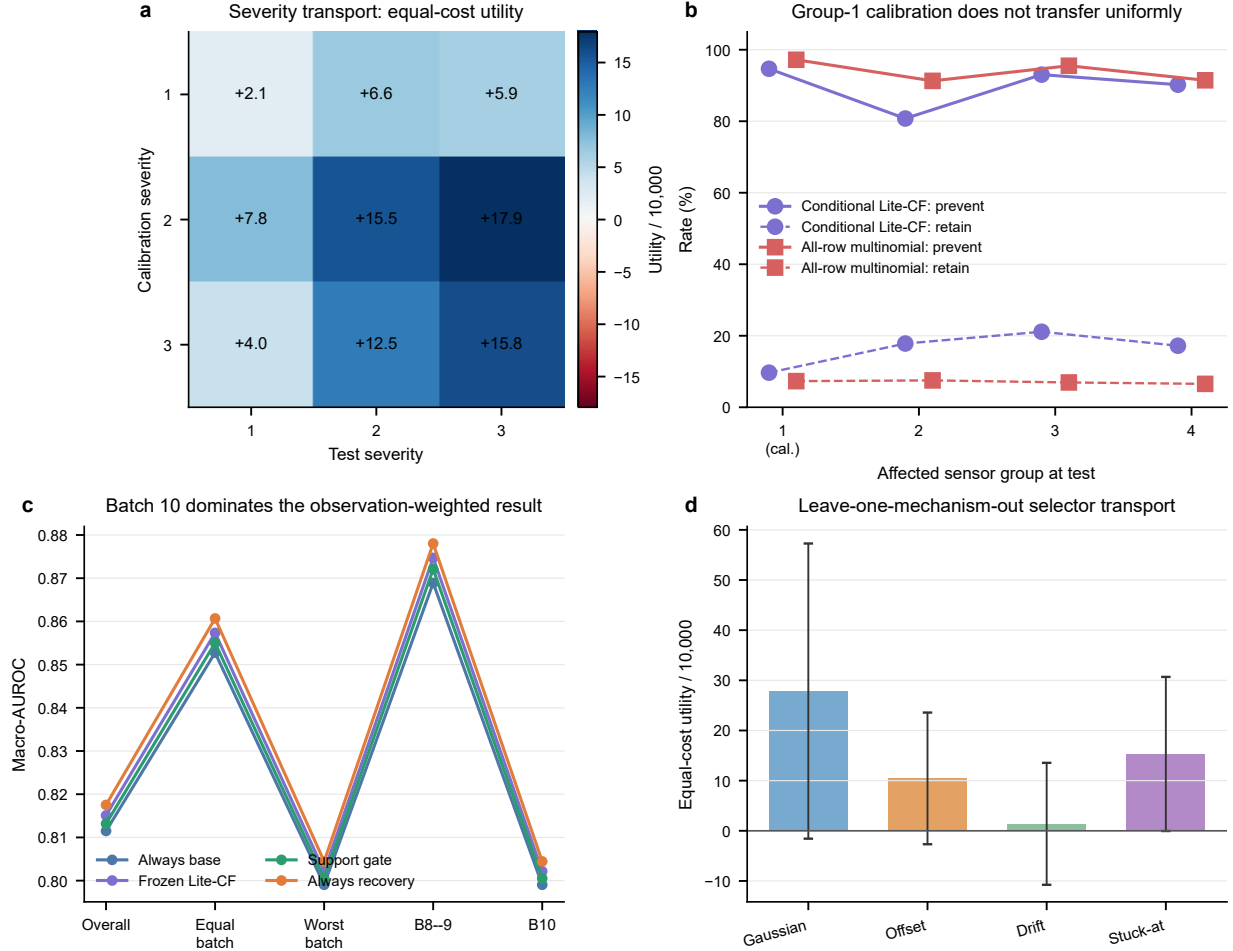


Figure 4: **Selector transport and acquisition-batch robustness.** **a**, conditional-router equal-cost utility for calibration and test severities 1–3. **b**, prevention (solid) and retention (dashed) after calibrating on sensor group 1 and testing groups 1–4. **c**, macro-AUROC under observation-weighted, equal-batch, worst-batch and Batches 8–9/Batch 10 views. **d**, leave-one-mechanism-out conditional-router utility; bars are means and whiskers are 95% fitted-pair intervals across 10 endpoint fits on fixed test rows.

## 4.5 Sparse calibration limited selector stability

The formal Lite-CF calibration sets contained a mean of 1,861 rows, but only 89.7 were correctness disagreements: 36.1 favoured PDRF and 53.6 favoured RO-PDRF-Lite. Frozen thresholds averaged 0.710 and ranged from 0.55 to 0.90. Across 10 repeated grouped fits per endpoint pair, mean out-of-fold AUROC was 0.720 (standard deviation 0.138; range 0.327–0.954), and thresholds ranged from 0.525 to 0.90. Only 5/10 pairs passed the prospective support gate. The 94.7%/9.7% result remains the auditable ungated estimate, but the gate-enforced 96.2%/6.6% result is the deployment primary because unsupported fits are not permitted to route.

Coefficient signs were most stable for Lite confidence (94%), base confidence (89%) and confidence change (83%). Dominant-sign fractions for the entropy terms were 61–67%. Strong shrinkage traded away conditional discrimination: mean out-of-fold AUROC increased from 0.690 at  $C = 0.01$  to 0.717 at  $C = 0.05$ , 0.800 at  $C = 1$  and 0.811 at  $C = 5$ . The prespecified  $C = 0.05$  is therefore a conservative estimation choice, not the value that maximizes calibration preference AUROC.

Calibration-size sensitivity showed the same boundary. Using 25%, 50%, 75% and 100% of calibration groups produced pooled mean prevention of 88.7%, 91.6%, 92.4% and 93.9%, respectively. Retention changed from 12.1% to 10.0%, while macro-AUROC remained near 0.81. These are repeated-subsampling summaries and differ from the equal-weight 40-cell headline estimand.

Endpoint temperature scaling changed the conditional router: uncalibrated endpoint features gave preference AUROC 0.563, prevention 95.7%, retention 6.6% and utility +10.8, compared with 0.562, 94.7%, 9.7% and +15.8 after temperature scaling. Platt calibration of the conditional score reduced preference NLL from 0.694 to 0.689 and Brier score from 0.249 to 0.239 without changing decisions when the frozen threshold was monotonically mapped. Isotonic calibration introduced flat regions and changed decisions, increasing utility but reducing prevention to 82.1%; it is therefore a sensitivity, not a replacement. Probability calibration and hard-decision selection remain separate objectives [34, 39].

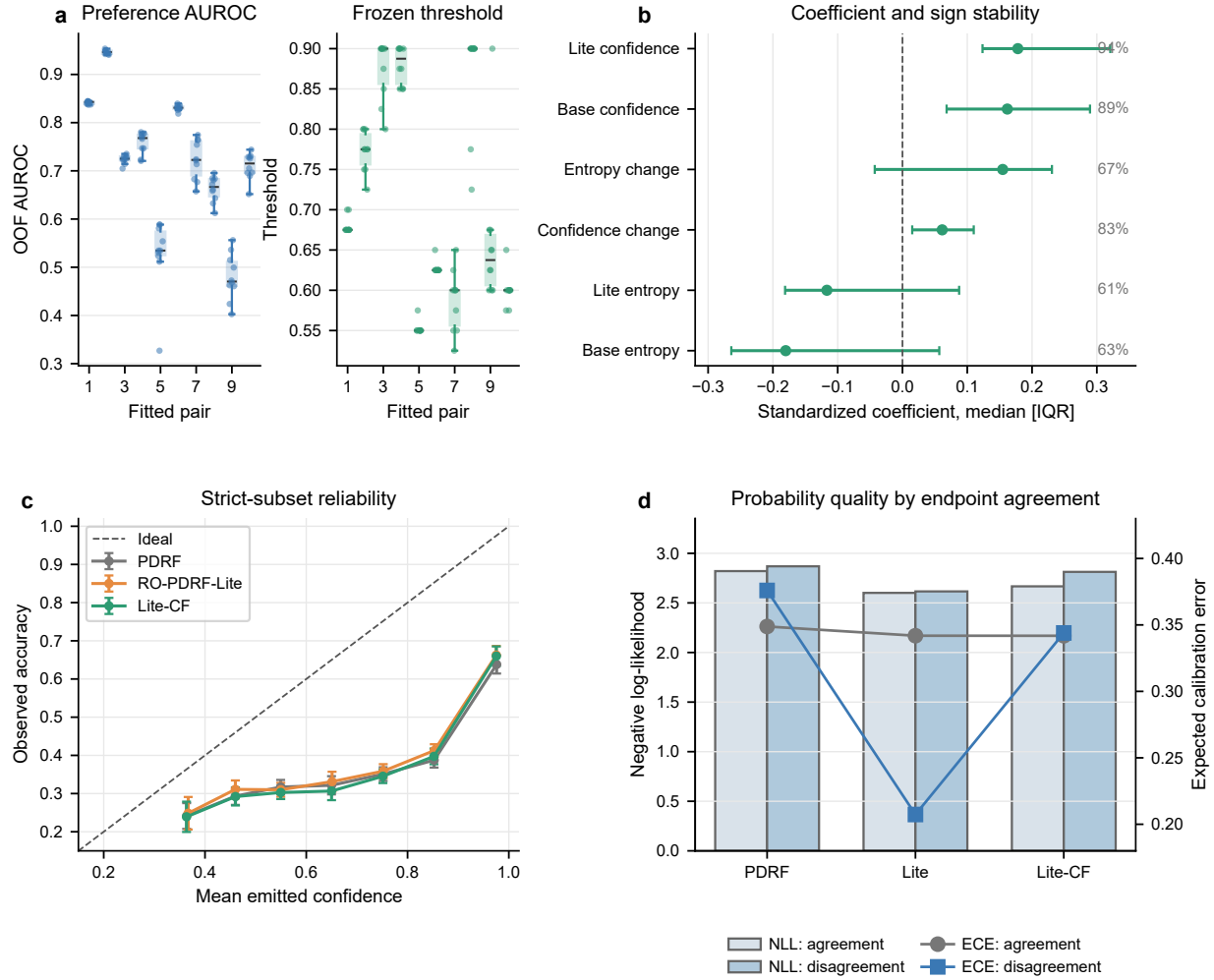


Figure 5: **Selector stability and final probability quality.** **a**, seed-wise out-of-fold conditional-preference AUROC and threshold distributions across 100 repeated grouped fits. **b**, median standardized Lite-CF coefficient with interquartile range; percentages are dominant-sign frequencies. **c**, reliability curves lie predominantly below the identity line, showing substantial overconfidence for all three outputs. **d**, negative log-likelihood and 15-bin expected calibration error after separating endpoint-class agreement from disagreement. Lite-CF emits one endpoint vector, not a mixture; score-calibration controls are reported separately.

## 4.6 Hydraulic single-rig sensitivity boundary

The guarded hydraulic analysis used four native component-condition tasks and controlled pressure- and vibration-group faults (Table 9). RO-PDRF-Full exceeded recovery-objective-matched RO-CAGF in six of eight task-fault conditions. The mean difference was +0.015, but the task-fault-seed hierarchical interval ( $-0.006$  to  $+0.036$ ) included zero. Performance ranged from near ceiling for cooler (RO-PDRF-Full macro-AUROC 0.996) to near chance for pump and accumulator (0.528 and 0.487), limiting architecture interpretation on the latter tasks.

The earlier random-stratified split yielded a larger mean difference (+0.044), indicating optimistic adjacent-cycle allocation. Guard bands reduced direct adjacency but did not create independent devices. No-interpolation and random-split results are therefore supplementary sensitivity evidence; the main result supports single-rig platform-internal robustness, not a general architecture ranking.

Table 9: **Hydraulic single-rig sensitivity analysis.** Guarded-block affected-subset macro-AUROC values are means  $\pm$  standard deviations across five matched seeds. The eight task-fault rows are conditions from one rig, not independent devices; the earlier random-stratified results are supplementary sensitivity evidence.

| Task        | Fault     | RO-CAGF           | RO-PDRF-Full      | Difference |
|-------------|-----------|-------------------|-------------------|------------|
| Cooler      | Pressure  | $0.942 \pm 0.023$ | $0.992 \pm 0.004$ | +0.050     |
| Cooler      | Vibration | $0.947 \pm 0.029$ | $1.000 \pm 0.000$ | +0.053     |
| Valve       | Pressure  | $0.730 \pm 0.026$ | $0.762 \pm 0.023$ | +0.032     |
| Valve       | Vibration | $0.739 \pm 0.029$ | $0.749 \pm 0.032$ | +0.010     |
| Pump        | Pressure  | $0.569 \pm 0.035$ | $0.536 \pm 0.018$ | -0.033     |
| Pump        | Vibration | $0.533 \pm 0.061$ | $0.521 \pm 0.022$ | -0.012     |
| Accumulator | Pressure  | $0.478 \pm 0.022$ | $0.491 \pm 0.023$ | +0.013     |
| Accumulator | Vibration | $0.474 \pm 0.047$ | $0.483 \pm 0.044$ | +0.009     |

## 4.7 AHU direct-diagnosis boundary

The chronological air-handling-unit tests contained 37,831 untouched future-time records, including 2,069 expert-annotated temperature-sensor faults. Because the target is the fault itself and no independent downstream state is available, this experiment cannot test whether suppressing that sensor recovers another prediction. Early fusion exceeded bounded recovery in all 15 building-seed pairs (mean AUROC difference  $-0.0341$  for bounded recovery,  $P < 0.001$ ), and leave-one-building-out bounded recovery collapsed to  $0.298 \pm 0.005$  in the hospital. The operational data therefore establish a direct-diagnosis and transfer boundary, not downstream recovery validation; complete subtype, shift, adaptation and risk analyses remain in the Supplementary Information.

## 4.8 Routed probability quality depended on endpoint agreement

Final routed probabilities were evaluated directly rather than inferred from the residual-harm rate. Reliability curves lay predominantly below the identity line, indicating substantial overconfidence for all methods. On strict endpoint-class agreements, Lite-CF NLL/ECE were 2.667/0.342, between PDRF (2.821/0.349) and RO-PDRF-Lite (2.601/0.342). On disagreements, Lite-CF NLL/ECE

were 2.813/0.344; unconditional Lite was better at 2.616/0.207, whereas PDRF was worse at 2.869/0.376. The software can expose the selected endpoint identity and conditional preference score with each probability vector. Downstream controllers should therefore treat confidence as endpoint-specific, not as one globally calibrated routed probability (Fig. 5c,d). Post-routing recalibration remains future work.

#### 4.9 Lite-CF reduced, but did not remove, deployment cost

PDRF and RO-PDRF-Lite each require one forward pass, 19,740 parameters and 38,528 counted linear-layer FLOPs per observation. Lite-CF requires two passes, 39,487 parameters and 77,056 endpoint FLOPs. With 12 intra-op threads, mean seed-level median batch-one latencies were  $0.602 \pm 0.263$  ms for PDRF,  $0.589 \pm 0.258$  ms for RO-PDRF-Lite and  $2.471 \pm 0.865$  ms for Lite-CF; complete-batch latencies were  $4.12 \pm 0.73$ ,  $3.65 \pm 0.66$  and  $10.21 \pm 2.09$  ms. At one intra-op thread, the corresponding batch-one values were  $0.273 \pm 0.090$ ,  $0.252 \pm 0.068$  and  $1.553 \pm 0.447$  ms, and complete-batch values were  $9.35 \pm 1.77$ ,  $5.17 \pm 0.58$  and  $11.60 \pm 0.81$  ms. All measurements used the same Intel Core i5-13500H, PyTorch 2.12.1+cpu, three warm-ups and seven timed calls, excluding loading. They do not establish embedded latency, energy efficiency or real-time readiness.

Event-normalized engineering cost was large. Across the 40 strict cells, the frozen router required 57,440 extra endpoint passes to produce 139 retained corrections and 91 net corrections: 413 extra passes per retained correction and 631 per net correction. It selected recovery 29,970 times, corresponding to 216 selected rows per retained correction and 329 per net correction. The support gate reduced total extra passes but also reduced net corrections, yielding 668 extra passes and 416 selected rows per net correction. For batch-one timing, the routed call exceeded the sum of separately timed endpoints by 1.03 ms at one thread and 1.28 ms at 12 threads; this residual includes feature construction, logistic scoring, array selection and Python orchestration. Full-batch residuals were non-additive ( $-2.93$  and  $+2.45$  ms), so they are reported as orchestration/timing residuals rather than pure router latency.

## 5 Discussion

The strict estimand and frozen-run discipline narrow the central interpretation. Clean-to-faulted distillation produced a useful recovery endpoint, but the bounded response architecture alone did not solve deployment choice. The stronger contribution is a decision protocol that distinguishes fault assignment from an intervention that reaches an available group, measures correction and error transfer, separates safety from utility and freezes every operating threshold before test evaluation. It should not be read as a general theory of sensor reliability.

Cross-fitting addresses threshold leakage, not selected-sample bias or sparse-event estimation. The original conditional score used only 89.7 correctness disagreements on average and therefore estimated preference conditional on disagreement. The all-row multinomial and two-stage controls target the required deployment population, but their rare base-better and recovery-better classes remained difficult. No test curve dominated the frozen point in both prevention and retention, while class-conditional and margin rules obtained higher equal-cost utility only by accepting substantially more harm. The estimand correction therefore changes what is measurable without creating a uniformly superior router.

The prospective support gate is central to deployment. Only 5/10 endpoint pairs passed, so the primary policy reverted the other half to PDRF. It achieved 96.2% prevention and 6.6% retention, but its fitted-pair utility interval crossed zero and worst-batch utility was  $-2.5$  per 10,000. A site should not transfer coefficients or thresholds from another fitted pair; it should use the base



endpoint until local event counts, repeated discrimination and threshold stability meet the declared gate.

Utility-aware thresholds clarify when selection adds value. At equal event values, unconditional recovery had the largest utility. At a two-to-one harm ratio, two-stage routing outperformed both endpoints; at ratios of five and ten, always PDRF dominated every learned policy. No single “safe” threshold is therefore application-independent. The values  $V_C$ ,  $C_H$  and  $\lambda$  must be defined in operational units—for example, avoided inspection cost, cost of a wrong control decision and compute/latency cost converted to the same per-10,000-observation scale—before calibration. A monotone score calibration cannot substitute for that value specification.

The two-pass design avoids leave-one-group-out inference but stores two 19,740-parameter endpoints (154.25 KiB in float32) and evaluates both. Pooled over the strict audit, frozen Lite-CF required approximately 413 extra passes per retained correction and 631 per net correction; about 329 recovery selections were required per net correction. The support gate required 668 extra passes and 416 selected rows per net correction because unsupported fits contributed no routed gains. Batch-one orchestration residual beyond the two separately timed endpoints was 1.03 ms at one thread and 1.28 ms at 12 threads; the full-batch residual ranged from  $-2.93$  to  $+2.45$  ms because independent timings are non-additive. These workstation measurements are not embedded latency or energy guarantees.

Mechanism, severity, affected-group and acquisition-batch transport all changed the decision balance. Leave-one-mechanism-out fitting intervals included zero; group-2 prevention fell to 80.8%; Batch 10 supplied 82.2% of strict observations and pulled the observation-weighted AUROC toward its lower value. These analyses are more informative than one aggregate mean, but they still reuse one chemical instrument and fixed rows. The fitted-pair intervals quantify optimization variation, not sampling from devices or natural faults.

Transfer tests define boundaries rather than confirm field recovery. The hydraulic result is single-rig sensitivity, and its guarded architecture interval crosses zero. The AHU target is the sensor fault itself, so diagnosis may reward signatures that recovery suppresses. A prospective study must include multiple independently operated devices, maintenance-adjudicated sensor failures and downstream targets that do not encode those failures. Endpoint fitting and site-specific calibration must precede outcome analysis, with the support gate, PDRF fallback and utility values frozen in advance.

Further boundaries remain. Routed probabilities are overconfident and endpoint-specific; no tested output dominates safety, ranking and calibration. Response saturation prevents device-health interpretation, and erroneous clean teachers or quality metadata can transfer errors. Evidence remains limited to controlled feature corruption, one chemical instrument, one hydraulic rig and expert/rule-based AHU labels. Multi-device prospective evaluation is required before field-recovery claims.

## 6 Conclusion

Support-gated selective recovery converted a useful but fallible recovery endpoint into an explicit, auditable deployment policy. The frozen ungated router prevented 94.7% of negative transfers while retaining 9.7% of corrections. The prospective policy admitted only 5/10 fitted pairs and reverted the remainder to PDRF, yielding 96.2% prevention and 6.6% retention. All-row and two-stage routers corrected the conditional-estimand mismatch and added utility at a two-to-one harm ratio, but always recovery was preferred at equal event values and always PDRF dominated at ratios of five or ten. Severity, sensor-group, unseen-mechanism and batch audits showed that thresholds did

not transport uniformly. Routed probabilities remained endpoint-specific and overconfident. The contribution is a reproducible risk-control and evaluation protocol for controlled available-group corruption, not evidence for calibrated device health, natural-failure recovery or device-independent field performance.

## Data and code availability

The Gas Sensor Array Drift Dataset (DOI: <https://doi.org/10.24432/C5RP6W>) and Condition Monitoring of Hydraulic Systems dataset (DOI: <https://doi.org/10.24432/C5CW21>) are available from the UCI Machine Learning Repository under CC BY 4.0. The operational AHU data are available from Figshare under CC BY 4.0 (DOI: <https://doi.org/10.6084/m9.figshare.27147678.v3>). Code, fixed seeds, environment information, prediction-level source data and figure/table scripts are publicly available in the `cee_revision` directory at <https://github.com/Corey9999/recovery-supervision-calibrated-fallback>. The exact computational snapshot for the all-row, utility, transport and batch audits is commit `f8ef1a17bdfc98377334dd9aa6903702cdc789c1`; the versioned submission release is `v1.4.0-major-router`. Exact dependencies are recorded in `requirements-lock.txt` and `environment.yml`, and the script-to-result manifest is Supplementary Table S1. A permanent software DOI has not been assigned.

## Ethics statement

The study used public engineering datasets and controlled computer-generated sensor corruptions. It involved no human participants, personal data, animals or biological specimens; institutional ethics review was not applicable.

## Author contributions

Riyang Luo: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing—original draft, Writing—review and editing, and Project administration.

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## Declaration of interests

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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